USER MANUAL

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TABLE OF CONTENTS

INTRODUCTION

aiSysMet is an AI-powered platform for analysis of metabolomics data and integration of multi-omics data. It leverages advanced biomedical data analytics to accelerate biomarker discovery. In this user manual, we describe various components of aiSysMet outlined in the following categories: management utilities, data import modules, data processing modules (MetCraft), metabolite annotation modules (MetaboQuest), image analysis modules (ImgCraft), data/integrative analysis modules (IntSys), and pipeline builder. Furthermore, demo datasets and projects are described.

MANAGEMENT UTILITIES

- The following management utilities allow users to create and manage cloud data storage and projects.

Data Manager

- Manages cloud storage space
- Allows users to upload and store raw and unprocessed omics data
- Facilitates efficient workflow execution by ensuring data accessibility from low-latency storage. Repetitive uploading of large raw data files from local storage degrades performance.
- Offers user-friendly interface for uploading, renaming, deleting, decompressing and downloading files and directories
- Operates with an intuitive, operative, and system-style interface similar to Google Drive

To upload an entire directory of data into the cloud, use the Data Manager as follows

- Log in to aiSysMet
- Click on the “Data Manager” button
- Select “Upload Directory” to copy/upload the entire content in the folder
- Select the directory on your local drive
- Click “Upload”

To upload selected files into the cloud, use the Data Manager as follows

- Log in to aiSysMet
- Click on the “Data Manager” button
    - Select “Upload Files” to copy/upload selected files
    - Select the files to upload on your local drive
- Click “Upload”

Project Manager

- Organizes data in the user space (uploaded data or pipeline generated outputs) on a project basis for easy management and retrieval.
- Facilitates collaboration among researchers by sharing projects/pipelines, original data, and results in a transparent manner.

To create a new project

- Log in to aiSysMet
- Click "Project Manager"
- Click "New Project"
- Enter a name for the project
- Build a pipeline by dragging and dropping modules from the left pane to the pipeline canvas
- Save the project by clicking "Save Project" on the right panel.

To load an existing project

- Log in to aiSysMet
- Click "Project Manager"
- Click "Load Project"
- Save the project by clicking "Save Project" on the right panel if you made changes to the project.

To load just the pipeline only from an existing project by excluding the corresponding project

- Log in to aiSysMet
- Click "Project Manager"
- Click "Load Pipeline"
- Note: If you save this project without loading data, only the pipeline will be saved

PIPELINE BUILDER

- aiSysMet allows users to build data analytics pipelines that can be saved as a project and can be retrieved into the pipeline canvas through the Project Manager.
- Users can drag modules from the left pane to the pipeline canvas by connecting the modules according to the desired workflow or sequence of analysis.
- Each module can be configured and run one at a time, or a pre-configured pipeline consisting of multiple modules can be run as a project at once.
- Users can modify existing pre-composed pipelines in case they want to re-evaluate their previous results.

To build a pipeline/project

- Click **Project Manager** to create a project. Then, load the project into the pipeline canvas.
- Starting with one of the Data Import modules, drag modules from the left pane to the pipeline canvas by connecting the modules according to the desired workflow or sequence of analysis.
- Configure each module by clicking first the module itself and then clicking **Configure Module** in the right pane.
- Execute individual modules by clicking **Run Module**. You can see the output by clicking **View Result** or download the result table by clicking **Download Result**.
- Click **Run Project** to run all modules in the project. This requires saving the project first.

- Click on **Save Project or Save Project As...** to save the project to be able to use it later by accessing it via the Project Manager.

Notes on Pipeline Builder

- The components that cannot be inserted or appended to the current pipeline are grayed. Through this, the pipeline builder ensures that the composition of the pipeline follows a logical workflow. Therefore, the user should observe the proper sequence for bringing components into the pipeline canvas
- After placing a module in the pipeline canvas, it can be configured with the appropriate processing settings before execution. To do this, use the Configure Module button. If the module is not configured, it will use its default settings upon execution.
- After configuring a module, click the Run Module button to execute the module.
- If you choose to run all modules within a project, click the **Run Project** button.
- The progress window below the pipeline canvas shows the current operations, selections, and the status of the operations, if available.
- The module execution status can also be determined using a color code (yellow indicates the module is running, pink indicates that an error has occurred, green means the module completed processing, results can be viewed or downloaded).
- Click the Delete **Module** button to remove an unwanted module from the pipeline canvas.
- Click the Save Project or Save Project As ... buttons to save the entire pipeline on the pipeline canvas
- Click **Clear Pipeline Canvas** to clear any existing pipelines in the pipeline canvas

AI ASSISTANT

- A RAG-powered feature that provides answers to user questions related to aiSysMet by referencing its system documentation and user manuals as an internal knowledge base.
- Documentation is converted into embeddings and stored in a vector database.
- Users can query the AI Assistant questions such as steps to build a pipeline, configure a certain module, view results, etc., in plain English by typing and submitting their question using the prompt box provided.
- The RAG model searches the vector database for a contextual answer to the query and generates a human-like response using OpenAI's LLM.
- It is important to note that because the Assistant's response is not generic information produced by the LLM, it does not provide satisfactory answers for questions outside the scope and context of our software.

MODULES

aiSysMet's modules are grouped as outlined below. Please note that all modules listed below are available to the user with a full subscription of aiSysMet. In addition to the Data Import modules, users can choose to subscribe to a subset of the three major modules (MetCraft, MetaboQuest, and IntSys) to be included in aiSysMet.

Data Import Modules

This group consists of modules for uploading data from local and cloud storage or retrieving data from pre-specified databases.

Data Upload

- Allows users to upload raw metabolomics data for data processing, processed metabolomics data for metabolite annotation, and any processed omics data for biomarker discovery from local or cloud storage
  - Raw/unprocessed metabolomics LC-MS/MS data in mzXML or mzML formats can be uploaded along with a list of precursor m/z values for metabolite annotation, group/sample labels for marker selection, and/or batch designations, if applicable.
  - Processed metabolomics data including MS/MS, MS1, or GC-MS in plain text format as well as a list of m/z values in either plain text or csv formats for metabolite annotation. In addition, processed metabolomics data can be entered directly into the provided window interface.
  - Processed omics data along with additional files including sample labels or group information, designation of batches, precursor m/z list, etc. for marker selection.
- Automatically identifies data types to determine which subsequent modules can be used to build pipelines

To load raw mzXML or mzML data:

- Drag the “Data Upload” module to the pipeline canvas
- Click “Configure Module”
- Choose “Cloud” to import data stored in the Data Manager or choose “Local” to import data from a local drive
- Select the mzXML/mzML files to upload
- Additional data may be added before pressing “Submit” including
  - “Precursor File” consisting of m/z and RT values to guide extraction of MS/MS data from the mzXML/mzML files
  - “Group Label” consisting of group label for each sample to help with subsequent differential analysis
  - “Batch Label” to perform batch correction
- Click “Submit”
- Click “Run Module”

To load processed mass spectrometric data for metabolite annotation:

- Drag the “Data Upload” module to the pipeline canvas
- Click “Configure Module”
- Choose “Cloud” to import data stored in the Data Manager or choose “Local” to import data from a local drive
- Choose the data type (m/z list, MS/MS, or MS1 data)
  - Note that you can upload more than one data type
- Select the data to import
- Click “Submit”

- Click “Run Module”

To load processed multi-omics data for differential or integrative analysis:

- Drag the “Data Upload” module to the pipeline canvas
- Click “Configure Module”
- Choose “Cloud” to import data stored in the Data Manager or choose “Local” to import data from a local drive
- Click “Upload Data”
- Select the data to import and click “Confirm”
- Click “Upload Group Label”
- Select the data to import and click "Confirm"
- You may upload multiple omics data by selecting “Add Another Omics Data”
- Click “Submit” when done
- Click “Run Module”

CRDC Data Retrieval

- Searches for preprocessed data from the NCI Cancer Research Data Commons via FireBrowse
- Users can specify various search criteria including:
  - Program (TCGA or CPTAC)
  - Primary site (breast, liver, ovarian, lung, brain, etc.)
  - Disease type
  - Omics data (mRNA-seq, miRNA-seq, proteomics, phosphoproteomics, etc.)
  - Imaging data
  - Grouping features (based on sample annotations such as disease, age, race, days-to-death, etc.)
- Displays available cohorts that meet user-defined selection criteria
- Displays a summary of imported data including demographics of study subjects

To retrieve omics data from the Cancer Research Data Commons (CRDC):

- Drag the “CRDC Data Retrieval” module to the pipeline canvas
- Click “Configure Module”
  - Select the Program (TCGA, CPTAC, TCIA)
  - Select the Primary site
  - Select the Disease type
- Click “Retrieve”
  - Select the project from the drop-down menu
  - Select the omic data type (mRNA-seq, miRNA-seq, proteomics, phosphoproteomics, etc.)
  - Select the grouping features based on sample annotations (e.g., disease, age, race, days-to-death, etc.)
- Click “Submit” when done
- Click “Run Module”

CRDC Image Retrieval

- Searches for raw images in the NCI Cancer Research Data Commons (CRDC)
- Users can specify various search criteria including:
    - Program (TCGA, CPTAC, TCIA)
    - Primary site (breast, liver, ovarian, lung, brain, etc.)
    - Images for selected subjects from the Collection
- Displays available cohorts that meet user-defined selection criteria
- Displays a summary of imported data including demographics of study subjects

To retrieve imaging data from CRDC

- Drag the "CRDC Image Retrieval" module to the pipeline canvas
    - Click "Configure Module"
    - Select the Modality (Slide Microscopy, MR, CT)
    - Select Program (TCGA, CPTAC)
    - Select Cancer Type
    - Select the Primary Site Location
    - Select Access Type
    - Select Collections
    - Click "Next" when done
    - Select the Collection
    - Use the Clinical Filters to select a subset of subjects based on ethnicity, pathologic stage, race, sex, viral status, age range, and BMI
    - Select the Patient IDs of the subjects
    - Press "Next"
    - Select the images to retrieve by selecting the Patient IDs
- Click "Submit"
- Click "Run Module"

Metabolomics Workbench Data Retrieval

- Allows users to import data from Metabolomics Workbench by selecting from a pre-specified list of Study IDs

To retrieve data from the Metabolomics Workbench

- Drag the "MetabolomicsWB Data Retrieval" module to the pipeline canvas
- Click "Configure Module"
    - Select the Study ID from the list provided
- Click "Submit" when done
- Click "Run Module"

MetCraft Modules - Data Processing

This group consists of the following modules to process raw LC-MS/MS metabolomics data or apply various data treatment methods to any processed omics or imaging data.

Peak Detection

- Analyzes unprocessed metabolomics data in mzXML or mzML formats to perform peak detection including peak picking, peak integration, and peak alignment
- Detects ion signals based on signal to noise ratio
- Reconstructs peak shapes using cubic spline interpolation

To perform peak detection, make sure that the mzXML/mzML files have been successfully uploaded via "Data Upload"

- Drag the "Peak Detection" module to the pipeline canvas and place it after the "Data Upload" module
- Click "Configure Module"
- Enter the desired ppm for peak detection
- Click "Submit"
- Click "Run Module"
- After the module turns green, the detected peaks can be viewed by clicking "View Result" or downloaded by clicking "Download Result"

Adduct/Isotope Recognition

- Facilitates subsequent metabolite annotation steps by pre-annotation, i.e., recognizing adducts and isotopes through peak clustering. This is important because one analyte may generate multiple peaks with distinct m/z values due to the effects of isotopes, adducts and neutral-loss fragments.

To perform adduct/isotope recognition, make sure that the mzXML/mzML files have been successfully uploaded via "Data Upload" and peak detection is performed by the "Peak Detection" module

- Drag the "Adduct/Isotope Recognition" module to the pipeline canvas and place it after the "Peak Detection" module
- Click "Configure Module"
- Enter the desired ppm, ionization mode, and adduct information
- Click "Submit"
- Click "Run Module"
- After the module turns green, the adducts and isotopes assigned to the peaks can be viewed by clicking "View Result" or downloaded by clicking "Download Result"

Data Filter

- Allows users to select a subset of features for subsequent analyses

- Features are removed based on: (1) a user-specified threshold for a coefficient of variation across all selected subjects; and (2) a threshold for the percentage of missing values

To select a subset of features from a processed dataset

- Drag the "Data Filter" module to the pipeline canvas and place it after a module that consists of a processed data matrix
- Click "Configure Module"
- Enter the minimum coefficient of variation allowable across all samples
- Enter the minimum percentage of samples the feature has to have a value
- Click "Submit"
- Click "Run Module"
- After the module turns green, the new data matrix size can be viewed by clicking "View Result" or the data matrix can be downloaded by clicking "Download Result"

Outlier Screening

- Applies Principal Component Analysis (PCA) to visualize samples that look different from the majority
- Identifies outliers that should be excluded in subsequent analyses

To detect outliers (samples that look different from the rest) based on a processed dataset

- Drag the "Outlier Screening" module to the pipeline canvas and place it after a module that consists of a processed data matrix
- Click "Configure Module"
    - Enter a threshold to define which samples will be considered as outliers
- Click "Submit"
- Click "Run Module"
- After the module turns green, a PCA plot that depicts the outliers can be viewed by clicking "View Result"

Missing Value Imputation

- Uses methods such as mean value, integer, k-nearest neighbor (KNN), and Random Forest (RF) to impute missing values, such as a peak missing in a small subset of samples but present in most.

To perform missing value imputation in a processed dataset

- Drag the "Missing Value Imputation" module to the pipeline canvas and place it after a module that consists of a processed data matrix
- Click "Configure Module"
    - Choose one of the methods (KNN sample wise, KNN feature wise, Mean Method, or Integer Method)
- Click "Submit"

- Click “Run Module”
- After the module turns green, the new data matrix after imputation can be viewed by clicking “View Result” or downloaded by clicking “Download Result”

Normalization

- This module provides access to several data normalization methods, including quantile normalization, median normalization, mean normalization, CycLoess, global robust linear regression (RLR), and global intensity normalization.

To perform data normalization on a processed dataset

- Drag the “Normalization" module to the pipeline canvas and place it after a module that consists of a processed data matrix
- Click “Configure Module”
    - Choose one of the methods (Quantile, Mean, Median, etc.)
- Click “Submit”
- Click “Run Module”
- After the module turns green, the new data matrix after normalization can be viewed by clicking “View Result” or downloaded by clicking “Download Result”

Batch Correction

- Uses empirical Bayes frameworks to adjust data from large-scale studies affected by run order or batch acquisition

To perform batch correction on a processed dataset, make sure that the batch label file is uploaded via the “Data Upload” module along with the original data

- Drag the “Batch Correction" module to the pipeline canvas and place it after a module that consists of a processed data matrix
- Click “Submit”
- Click “Run Module”
- After the module turns green, the new data matrix after batch correction can be viewed by clicking “View Result” or downloaded by clicking “Download Result”

MetaboQuest Modules - Metabolite Annotation

This group includes several modules that perform mass-based search and spectral matching for metabolite annotation. Other modules that apply Isotopic pattern analysis, network-based annotation, IF-THEN rule, and compound fingerprint prediction help organize and rank putative metabolite IDs.

Spectral Matching

- Searches for putative metabolite IDs by matching MS/MS or EI-MS spectra with our spectral database (SpectDB)

To perform spectral matching based on MS/MS or EI-MS data, make sure that the data are uploaded via the "Data Upload" module in plain text format or in mzXML/mzML format along with a precursor file in .csv format.

- Drag the "Spectral Matching" module to the pipeline canvas and place it after a module that consists of a processed data matrix
- Click "Configure Module"
  - Select the precursor mass tolerance in ppm, MS/MS spectrum tolerance in ppm, minimum score threshold, number of matching peaks, spectrometry method (MS/MS or GC-MS), ionization mode, experimental/in silico setting, adducts, and libraries to search against.
- Click "Submit"
- Click "Run Module"
- After the module turns green, the list of metabolites with spectra similar to the uploaded ones can be viewed along with their corresponding matching scores by clicking "View Result" or downloaded by clicking "Download Result"

Compound Fingerprint Prediction

- Uses a deep learning model to predict compound fingerprints based on MS/MS data
- Utilizes predicted fingerprints to rank candidate metabolites
- Designed for analytes that lack reference measurements in spectral libraries or have low spectral matching scores

To use the compound fingerprint approach for annotation, make sure you run the Spectral Matching module successfully first

- Drag the "Compound Fingerprint" module to the pipeline canvas and place it after Spectral Matching
- Click "Run Module"
- After the module turns green, the list of metabolites from spectral matching can be viewed along with their corresponding spectral matching and fingerprint scores by clicking "View Result" or downloaded by clicking "Download Result"

Mass-Based Search

- Enables search for putative metabolite IDs in MetDB based on m/z values
- Users are able to enter m/z values or use uploaded or processed data from a preceding module to search for putative IDs
- Calculates monoisotopic mass values based on the m/z values and user-specified adducts, ionization mode, mass tolerance in ppm.

To perform mass-based metabolite annotation, make sure you upload an m/z list in .csv format or use the Peak Detection module to obtain m/z values from mzXML/mzML files

- Drag the "Mass-Based Search" module to the pipeline canvas and place it after Data Upload if m/z list is provided or after "Peak Detection" or "Adduct/Isotope Recognition" modules

- Click “Configure Module”
    - Select the mass tolerance in ppm, the ionization mode, adducts
- Click “Run Module”
- After the module turns green, the list of metabolites from spectral matching can be viewed along with their corresponding spectral matching and fingerprint scores by clicking “View Result” or downloaded by clicking “Download Result”

IF-THEN Rule

- Allows users to select IF-THEN rules in order to combine, remove, or mark putative metabolite IDs.

To organize the putative IDs obtained from a mass-based search

- Drag the “IF-THEN Rule" module to the pipeline canvas and place it after the “Mass-Based Search” module
- Click “Configure Module”
    - Select the IF-THEN Rules by clicking the IF from the IF column and the THEN from the THEN column followed by clicking the “Add Rule” button. Multiple rules can be added.
- Once all rules are added, click “Submit Rules”
- Click “Run Module”
- After the module turns green, the list of putative IDs organized by the IF-THEN rules can be viewed by clicking “View Result” or downloaded by clicking “Download Result”

Isotopic Pattern Analysis

- Assigns scores to putative metabolite IDs based on their isotopic patterns
- Compares potential IDs with varying elemental formulas
- Calculates scores by comparing observed isotopic patterns from MS spectra with theoretical isotopic patterns

To assign scores to putative metabolite IDs based on their isotopic pattern, make sure that “Adduct/Isotope Recognition" module is successfully run prior to “Mass-Based Search” or the m/z list uploaded via the “Data Upload” module along with information on isotopes

- Drag the “Isotopic Pattern Analysis" module to the pipeline canvas and place it after the “Mass-Based Search” or “IF-THEN Rule” modules
- Click “Configure Module”
    - Indicate the columns that correspond to the first and last samples the user would like to consider for isotopic pattern analysis
- Click “Submit”
- Click “Run Module”
- After the module turns green, the list of putative IDs can be viewed along with isotopic pattern scores (for those who have isotopic peaks) by clicking “View Result” or downloaded by clicking “Download Result”

Network-Based Annotation

- Assigns scores to putative IDs using a network-based method
- Constructs a metabolic network by extracting biochemical pathway information from databases such as MetaCyc and KEGG
- Assigns probability scores to putative IDs, indicating the likelihood of their accuracy for a peak.

To use a network-based approach to assign scores to putative metabolite IDs obtained by mass-based search,

- Drag the "Network-Based Annotation" module to the pipeline canvas and place it after the "Mass-Based Search" or "IF-THEN Rule" modules
- Click "Configure Module"
    - Indicate if you prefer to display the network graph after processing
- Click "Submit"
- Click "Run Module"

After the module turns green, the list of putative IDs can be viewed along with scores from the network-based annotation method by clicking "View Result" or downloaded by clicking "Download Result"

ImgCraft modules - Imaging Data Analysis

The modules in this group enable users to process three imaging modalities (WSI - Whole Slide Imaging, CT - Computed Tomography, and MRI - Magnetic Resonance Imaging).

Image Import

- The module imports and prepares images selected in the CRDC Image Retrieval module for subsequent analysis.

To import images selected by the CRDC Image Retrieval module,

- Drag the "Image Import" module to the pipeline canvas and place it after the "CRDC Image Retrieval" module
- Click "Run Module"

WSI Analysis

- This module analyzes WSI data using foundation models that are fine-tuned to the specific organ to identify specific patches, lesions, or diagnostic features.

To perform WSI analysis,

- Drag the "WSI Analysis" module to the pipeline canvas and place it after the "Image Processing" module
    - Select the analysis
    - Select the analysis Content
- Click "Run Module"

CT Analysis

- This module performs segmentation of CT images using foundation models that are fine-tuned to the specific organ.

To perform CT analysis,

- Drag the "CT Analysis" module to the pipeline canvas and place it after the "Image Processing" module
    - Select the analysis Method
    - Select the analysis Content
- Click Submit
- Click Run Module

MRI Analysis

- This module performs segmentation of MR images using foundation models that are fine-tuned to the specific organ.

To perform MRI analysis,

- Drag the "MRI Analysis" module to the pipeline canvas and place it after the "Image Processing" module
    - Select the analysis Method
    - Select the analysis Content
- Click Submit
- Click Run Module

IntSys Modules - Integrative Data Analysis

The modules in this group allow users to identify significantly altered metabolites or multi-omics features through integrative analysis. Each module can be used for analysis of single omics or multi-omics data. Modules are linked to tools for visualization including ROC curves, Box plots, Volcano plots, Heatmaps, t-SNE, and Hierarchical clustering.

Survival Analysis

- Performs survival analysis using the Cox Proportional model or machine learning (XGBoost)

To perform survival analysis on imaging data imported along with survival information including survival time, event designation, and censored subjects,

- Drag the "Survival Analysis" module to the pipeline canvas and place it after "WSI Analysis", "CT Analysis", or "MRI Analysis" module
    - Select the model type (COX, XGBoost)
    - Select the Cross-Validation Folds
- Click "Submit"
- Click "Run Module"

After the module turns green, the survival analysis results can be viewed along with Kaplan-Meier curves.

Univariate Statistical Analysis

- Analyzes preprocessed single omics, multi-omics, or imaging data using parametric (Student t-test) or non-parametric (Mann-Whitney U-test) statistical methods to identify significantly altered features between two groups of samples.
- Analyzes matched/paired samples (i.e., tumor and adjacent non-tumors) using parametric (paired t-test) or non-parametric (Wilcoxon signed-rank test) to identify significantly altered features between two groups of samples
- Multi-omics and imaging features from the same subjects are simply concatenated for univariate analysis.

To perform significant analysis using univariate statistical methods, make sure that either: (1) processed single/multi-omics data uploaded with group labels are provided; or (2) mzXML/mzML data are uploaded with group labels and peak detection is performed. Before statistical analysis, various processing steps may be performed using modules such as "Data Filter", "Outlier Screening", "Missing Value Imputation", "Normalization", and "Batch Correction").

- Drag the "Univariate Statistical Analysis" module to the pipeline canvas and place it after the "Data Upload" module or any of the processing steps described above
- Click "Configure Module"
  - Select the desired statistical method (e.g., T-test, Paired T-test, Wilcoxon, Mann-Whitney U-test, Chi-Squared Test & T-Test)
- Click "Submit"
- Click "Run Module"

After the module turns green, the list of features can be viewed along with p-values, FDR, fold change by clicking "View Result" or downloaded by clicking "Download Result". Additionally, PCA plot, t-SNE cluster analysis result, heatmap, PLS-DA plot, volcano plots, box plots, and ROC curves can be viewed in the "Chart" tab of the result window.

Multivariate Regression Analysis

- Allows users to apply multivariate analysis (Lasso Regression and Elastic Net) to select a panel of disease-associated features.

To use multivariate regression for feature selection, make sure that either: (1) processed single/multi-omics data uploaded with group labels are provided; or (2) mzXML/mzML data are uploaded with group labels and peak detection is performed. Before statistical analysis, various processing steps may be performed using modules such as "Data Filter", "Outlier Screening", "Missing Value Imputation", "Normalization", and "Batch Correction").

- Drag the "Multivariate Regression" module to the pipeline canvas and place it after the "Data Upload" module or any of the processing steps described above
- Click "Configure Module"
  - Select the desired statistical method (e.g., Lasso Regression, Elastic Net)

- Click “Submit”
- Click “Run Module”

After the module turns green, the list of features can be viewed along with their regression coefficients ranked in the order of their relevance.

Network-Based Analysis

- Uses network-based methods for differential feature analysis of analytes in single omics, multi-omics, or imaging data
- Uses differential networks to compare correlations between analyte pairs within disease groups vs. control groups
- Helps users understand changes in pairwise interactions of analytes related to disease

To perform network-based analysis for feature selection, make sure that either: (1) processed single/multi-omics data uploaded with group labels are provided; or (2) mzXML/mzML data are uploaded with group labels and peak detection is performed. Before statistical analysis, various processing steps may be performed using modules such as “Data Filter”, “Outlier Screening”, “Missing Value Imputation”, “Normalization”, and “Batch Correction”).

- Drag the "Network-Based Analysis" module to the pipeline canvas and place it after the “Data Upload” module or any of the processing steps described above
- Click “Configure Module”
    - Select the rho value
- Click “Submit”
- Click “Run Module”

After the module turns green, the list of features can be viewed and ranked in the order of their activity scores

Machine Learning

- Uses two machine learning methods: support vector machine and random forest
- Uses the recursive feature elimination method, which selects disease-associated features from single or multi-omics data
- Standardized multi-omics features are combined into vectors for each sample to identify features that predict disease status

To use machine learning methods for feature selection, make sure that either: (1) processed single/multi-omics data uploaded with group labels are provided; or (2) mzXML/mzML data are uploaded with group labels and peak detection is performed. Before statistical analysis, various processing steps may be performed using modules such as “Data Filter”, “Outlier Screening”, “Missing Value Imputation”, “Normalization”, and “Batch Correction”).

- Drag the "Machine Learning" module to the pipeline canvas and place it after the “Data Upload” or any of the data processing modules as well as Univariate Statistical Analysis Module
- Click “Configure Module”

- Select the desired variable selection method (e.g., RFE with Support Vector Machine or Random Forest)

- Click "Submit"
- Click "Run Module"

After the module turns green, the list of features can be viewed and ranked in the order of their relevance.

DEMO DATA

MetCraft - Unprocessed Metabolomics Data

- **Demo1**: a folder consisting of: (1) Demo1a_mzML_pos: a folder of 8 mzML files acquired by metabolomics analysis of 8 QC samples using LC-MS/MS in the positive mode and a precursor file that indicates the m/z and RT values of all precursor ions expected for each mzML file. This demo dataset can be used for annotation of the analytes indicated in the precursor file using the Spectral Matching module. The module may extract the MS/MS spectra guided by the m/z provided in the precursor file. Users may choose to use the RT values in the precursor file or let the module automatically choose high quality MS/MS spectra across all scans. This demo dataset can also be used for peak detection using the Peak Detection module; and (2) Demo1b_mzML_neg: the same samples as Demo1a analyzed in the negative mode. Note that both the Peak Detection and Spectral Matching modules work as well with LC-MS/MS data in mzXML format.

MetaboQuest - Processed Data for Metabolite Annotation

- **Demo2**: a folder consisting of: (1) Demo2a_MSMS_pos: a folder of 12 files each consisting of an MS/MS spectrum acquired in the positive mode as well as a file listing the precursor m/z values corresponding to each of the 12 files; and (2) Demo2b_MSMS_pos.txt: all 12 MS/MS spectra from Demo2a listed in one file and a file listing the precursor m/z values corresponding to each MS/MS spectrum. These datasets can be used for metabolite annotation using the Spectral Matching module by uploading the 12 MS/MS spectra or one single file consisting of all MS/MS spectra along with the corresponding precursor files.
- **Demo3**: a folder consisting of: (1) Demo3a folder with two folders for MS1 spectra and MS/MS spectra acquired by LC-MS/MS in the positive mode as well as a file listing the precursor m/z values corresponding to the spectra; (2) Demo3b_MS1.txt that contains MS1 spectra for batch processing along with a file listing all the precursor m/z values; and (3) Demo3c_MS2.txt containing MS/MS spectra for batch processing along with a file listing all the precursor m/z values.
- **Demo4**: a folder consisting of: (1) Demo4a_EI.txt: a set of 5 EI spectra acquired by GC-MS. This demo dataset can be used for batch metabolite annotation using the Spectral Matching module by choosing the GC-MS platform; (2) Demo4b_EI.txt: the same datasets as Demo3a but combined in one file. This demo dataset can be used for

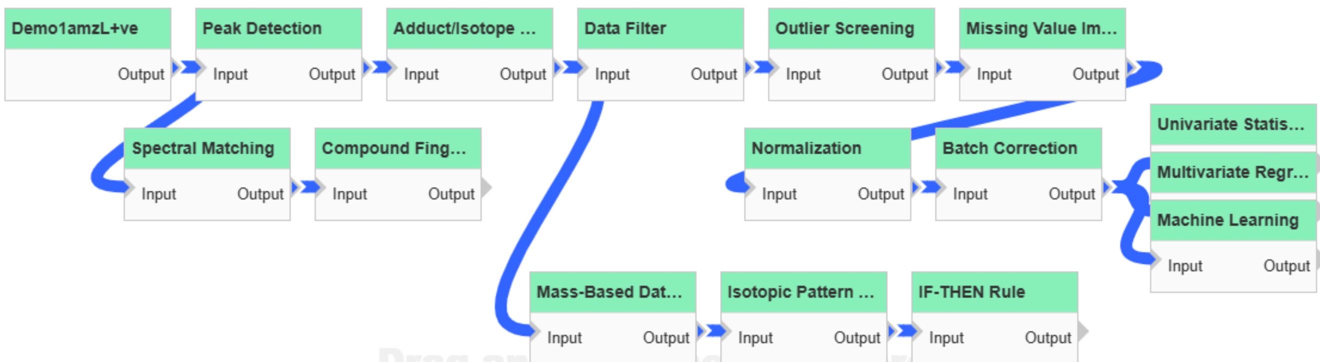
metabolite annotation using the Spectral Matching module by choosing the GC-MS platform.

IntSys - Processed Multi-Omics Data for Integrative Analysis

- **Demo5**: a folder consisting of three omics (metabolomics, glycomics, and proteomics) datasets acquired from the same set of samples. Each dataset, separately or in combination, can be used to test data processing and integrative analysis modules.
- **Demo6**: a folder consisting of three omics (mRNA expression profile, miRNA expression profile, and metabolomics profile) datasets acquired from the same set of samples. Each dataset, separately or in combination, can be used to test data processing and integrative analysis modules.
- **Demo7**: a folder consisting of two omics (mRNA expression profile and miRNA expression profile) datasets acquired from the same set of samples comprising tumor and non-tumor pairs. Each dataset, separately or in combination, can be used to test data processing and integrative analysis modules.
- **Demo8**: a folder consisting of preprocessed metabolomics data acquired in the positive mode, proteomics data, and glycomics data from an overlapping set of samples and three groups of annotation files. The datasets can be used to test data processing and integrative analysis modules.

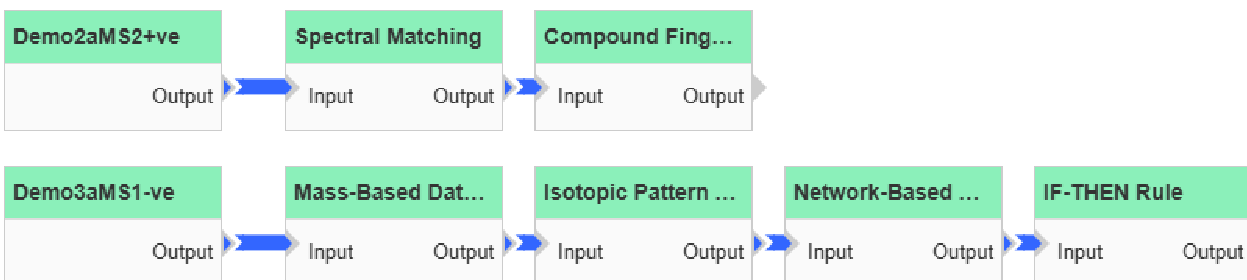
DEMO PIPELINES/PROJECTS

MetCraft - Data Processing



[Click here to watch a demo video for this project]

MetaboQuest - Metabolite Annotation



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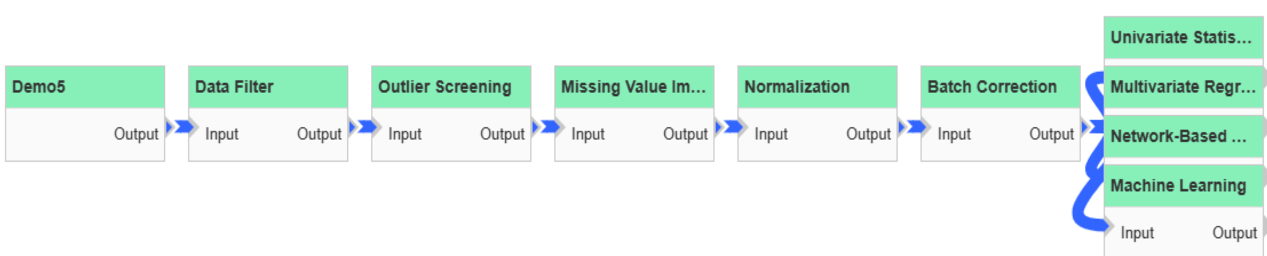
IntSys - Multi-Omics Data Integration



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